Production and growth (Topic 8/ Chapter 25)

How productivity is determined?

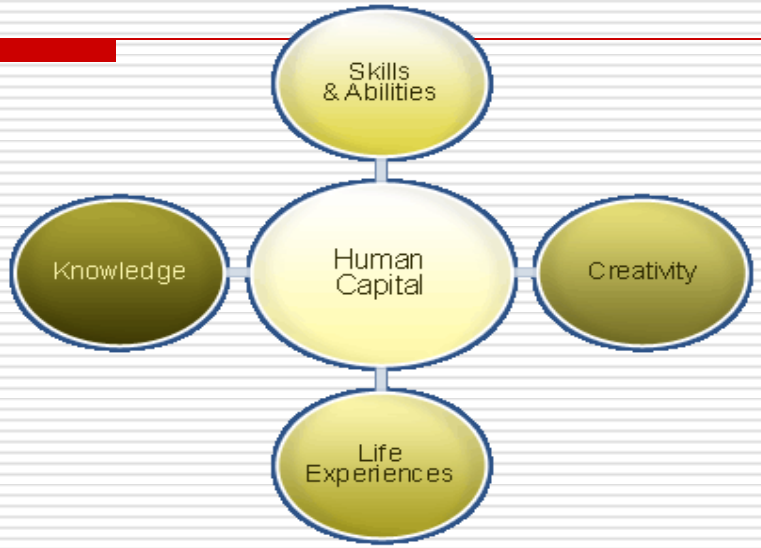
- The standard of living in an economy depends on the economy's ability to produce goods and services.
- Productivity, in turn, depends on the amount of physical capital, human capital, natural resources and technological knowledge available to workers.

Contd.

- Physical capital: The stock of equipment and structures that are used to produce goods and services is called physical capital or just capital.
- Human capital: The knowledge and skills that workers acquires through education, training and experience. Like physical capital human capital raises the economies ability to produce goods and services.
- Natural resources: The inputs into the production of goods and services that are provided by nature, such as rivers and mineral deposits.
- Technological knowledge: Society's understanding of the best ways to produce goods and services.

Contd.


Air
Water
Soil
Trees
Forest
Wildlife
Coal
Petroleum
Natural Gas
Minerals


Skills & Abilities
Knowledge
Human Capital
Creativity
Life Experiences


SOFTWARE
INFORMATION
TECHNOLOGIES
STORAGE
SCIENCE
INTERNET
PETABYTES
DATA
BIG
SOCIAL
WORLDWIDE
PROCESS
MILLION
NAS
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ANALYSIS
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Economic growth and public policy:

- Government policy can try to influence the economy's growth rate in many ways:
  - by encouraging savings and investment, encouraging investment from abroad,
  - fostering education,
  - maintaining property rights and political stability,
  - free trade,
  - promoting the research and development of new technologies and controlling population growth.

What can government policy do to raise productivity and living standards?

Government policy can try to influence the economy's growth rate in many ways. These are given below:

- Encouraging savings and investment
- Encouraging investment from abroad
- Fostering education
- Health and Nutrition
- Maintaining property rights
- Maintaining political stability
- Allowing free trade
- Promoting the research and development of new technologies
- Controlling population growth

Saving and Investment In The national Income Accounts

1. Consumption (C)
2. Investment (I)
3. Government purchases (G)
4. Net exports (NX)

- The circular flow diagram contains only first three components
- For a closed economy, first three are relevant, NX are zero.

What determines the demand for goods and services?

- For a closed economy:

  $Y = C + I + G$

-- Households consume some of the economy's output

-- Firms and households use some of the output for investment

-- The govt. buys some of the output for public purposes

- We want to see how GDP is allocated among these three uses.

Consumption (c)

- Disposable income: Y – T
- Households divide their disposable income between consumption and saving.
- We assume that the level of consumption depends directly on the level of disposable income
- C = C (Y-T)

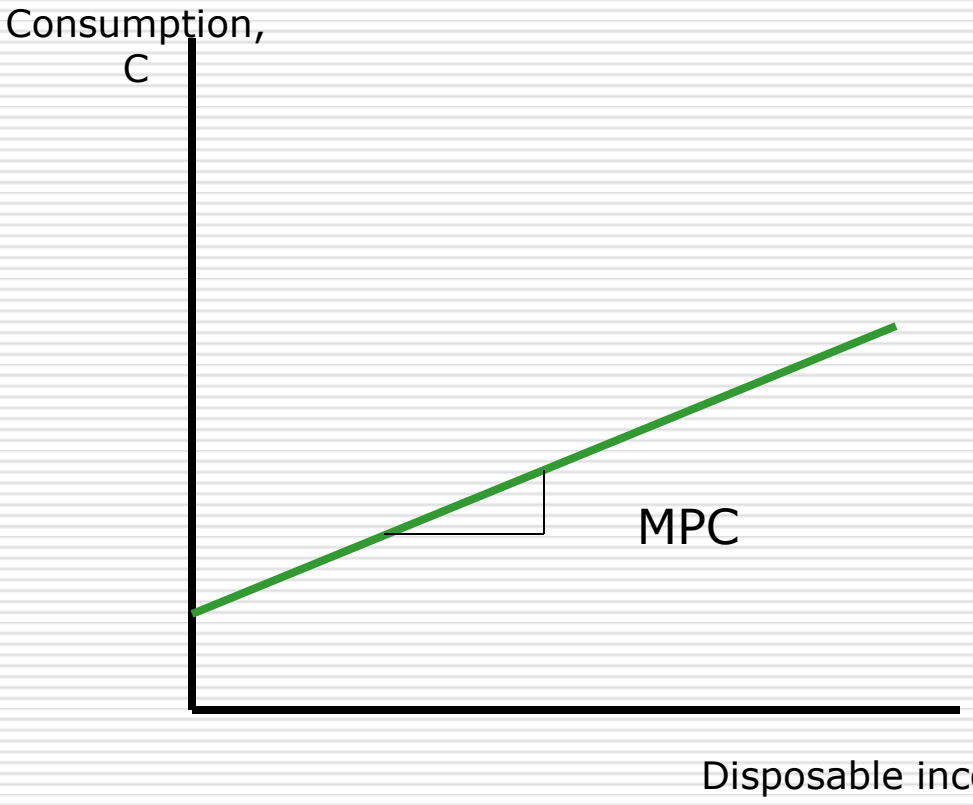
Consumption (C)

- Marginal Propensity to consumption (MPC):

  This implies the amount by which consumption changes when disposable income increases by one dollar (taka).

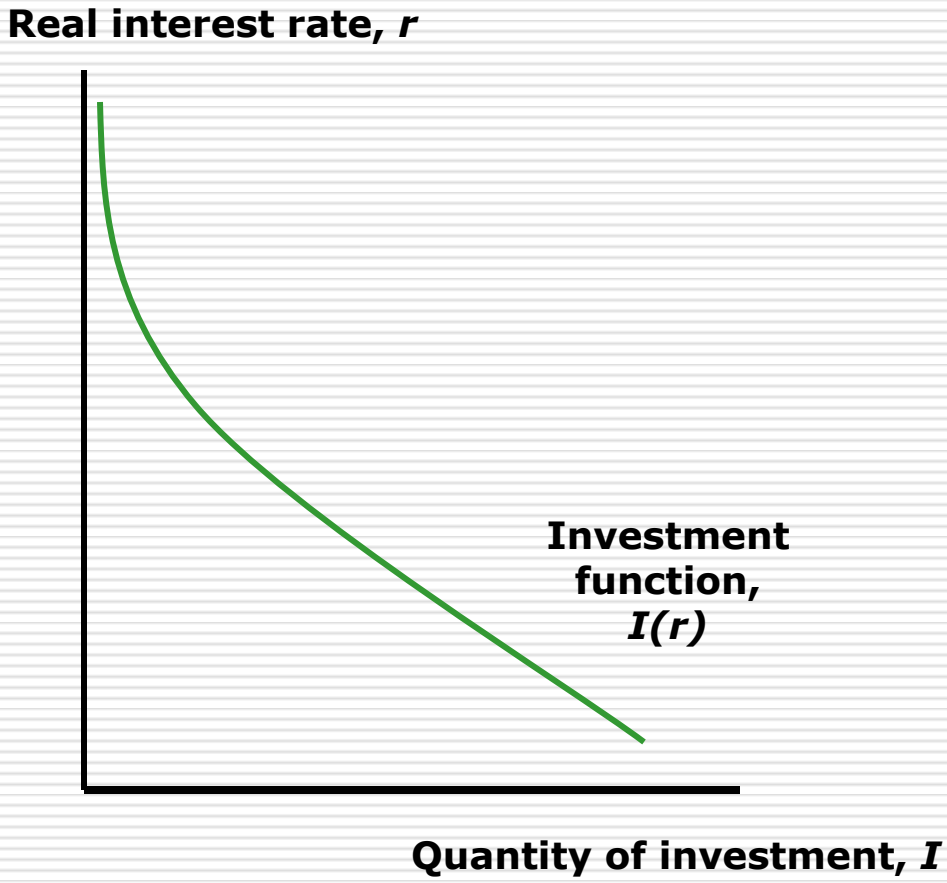
  $$0 < MPC < 1$$

Consumption Function (c)



The consumption function relates consumption C to disposable income Y – T. MPC is the amount by which consumption increases when disposable income increases by one taka.

Investment Function (I)



- The investment function relates the quantity of investment *I* to the real interest rate *r*.
- Investment depends on the real interest rate because the interest rate is the cost of borrowing.
- The investment function slopes downward. When the interest rate rises, fewer investment projects are profitable.

Government Purchases (G)

- G is the third component of D for good & services.
- Purchases for defense
- Services of government employees
- Spending on infrastructure development
- G = T Balance Budget
- G>T Budget deficit
- G < T Budget surplus

Saving & Investment In the National Income Accounts:

- National income accounting identities reveal some important relationships among macroeconomic variables. In particular for a closed economy, national saving must equals investment. *Financial institution* are the mechanism through which the economy matches one person's saving with another person's investment.

Equilibrium in the financial market

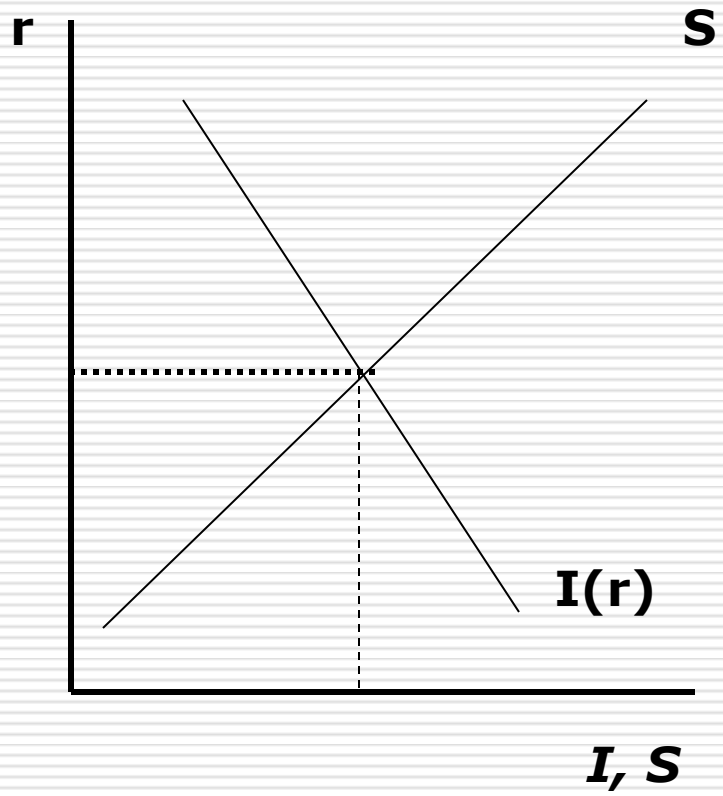
$Y - C - G = I$

$S = I$ (S← National saving)

S = Private saving + public saving

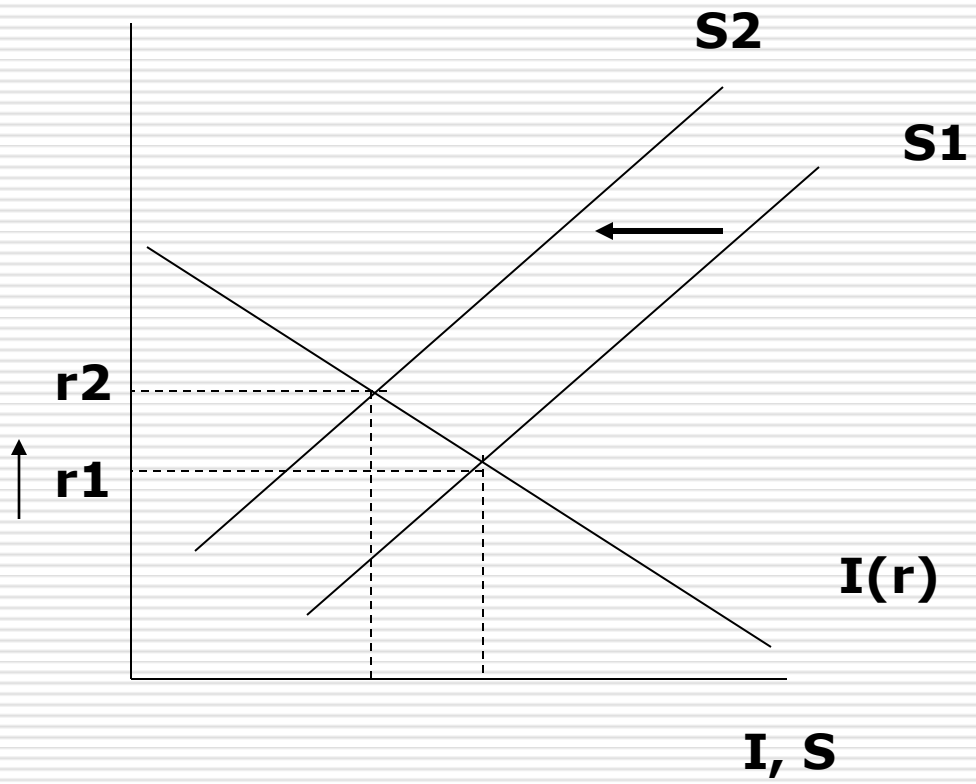
$= (Y - T - C) + (T - G) = I$

Equilibrium in the financial market



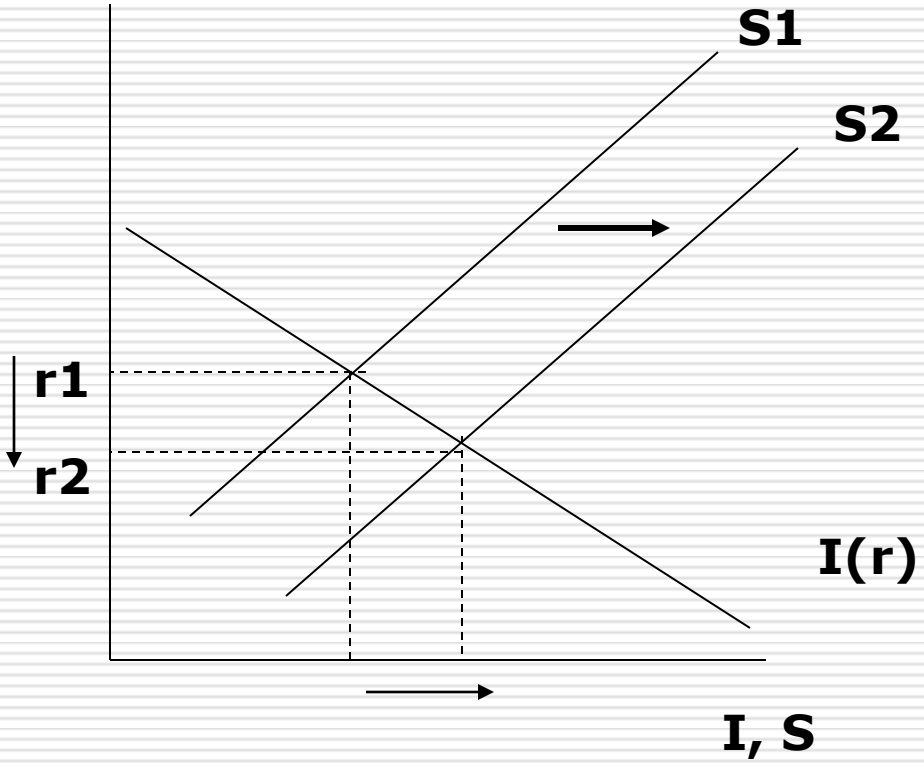
- The interest rate adjusts to bring saving and investment into balance. The upward line represents saving –*the supply of loanable funds*.
- The downward sloping line represents investment —*demand for loanable funds*.
- The interesction of these two lines determines the equilibrium interest rates.

Changes in saving: The effects of fiscal policy (G, T)



- The increase in G causes interest rate to increase and investment to decrease. Decrease in T will have the same effect
- When government budget deficit crowds out investment (i.e. fall in investment because of government borrowing is called crowing out), it reduces the growth of productivity and GDP.
- A decrease in tax raises disposable income by ΔT, consumption goes up by MPC x ΔT. National saving falls by the same amount as consumption rises. So, supply of loanable funds shift to the left, which increases the interest rate and crows out investment.

Changes in saving: The effects of fiscal policy (G, T)



- Government budget surpluses work just the opposite as budget deficit. Thus , a budget surplus (i.e. T > G) increases the supply of loan able funds, reduces the interest rate, and stimulates investment. Higher investment, in turn, means greater capital accumulation and more rapid economic growth.